Model Performance Analysis Report

1. Summary of Model Performance

The project evaluated two machine learning models to predict track popularity: **Linear Regression** and **Random Forest Regressor**. The performance was measured using R-squared (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

Metric	Linear Regression	Random Forest Regressor
R^2 Score	0.4497	0.4434
MAE	0.5826	0.5856
RMSE	0.7353	0.7396

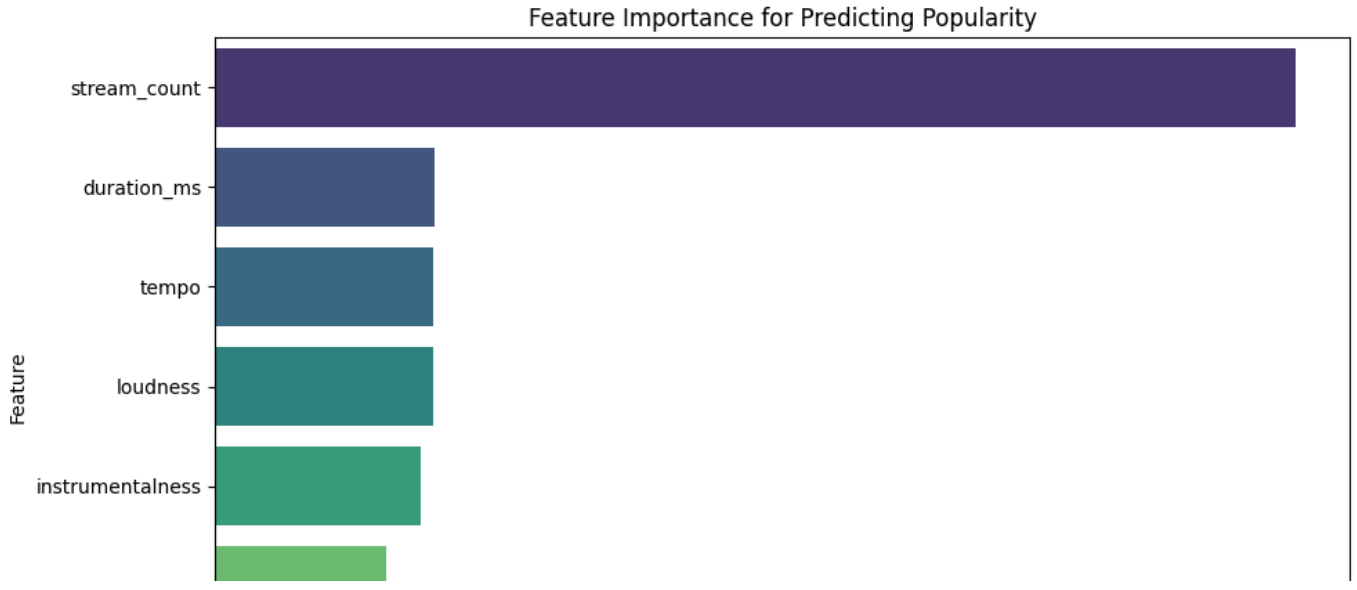
Both models showed similar performance, with **Linear Regression** slightly outperforming the Random Forest Regressor across all metrics. An R^2 score of approximately **0.45** suggests that the models can explain about 45% of the variance in the popularity of the tracks based on the provided features.

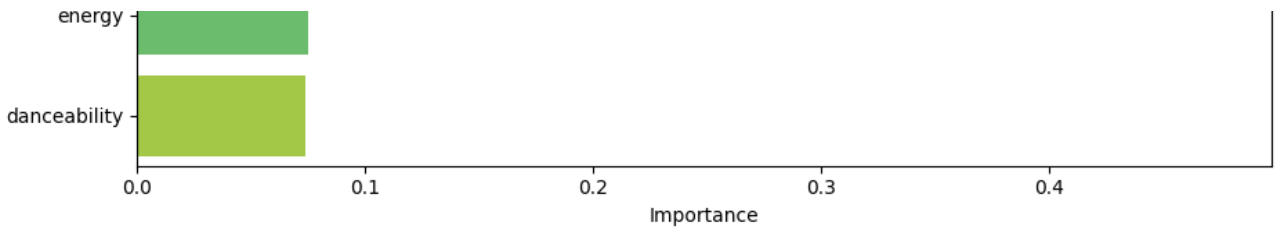
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2. Key Visualizations

Feature Importance

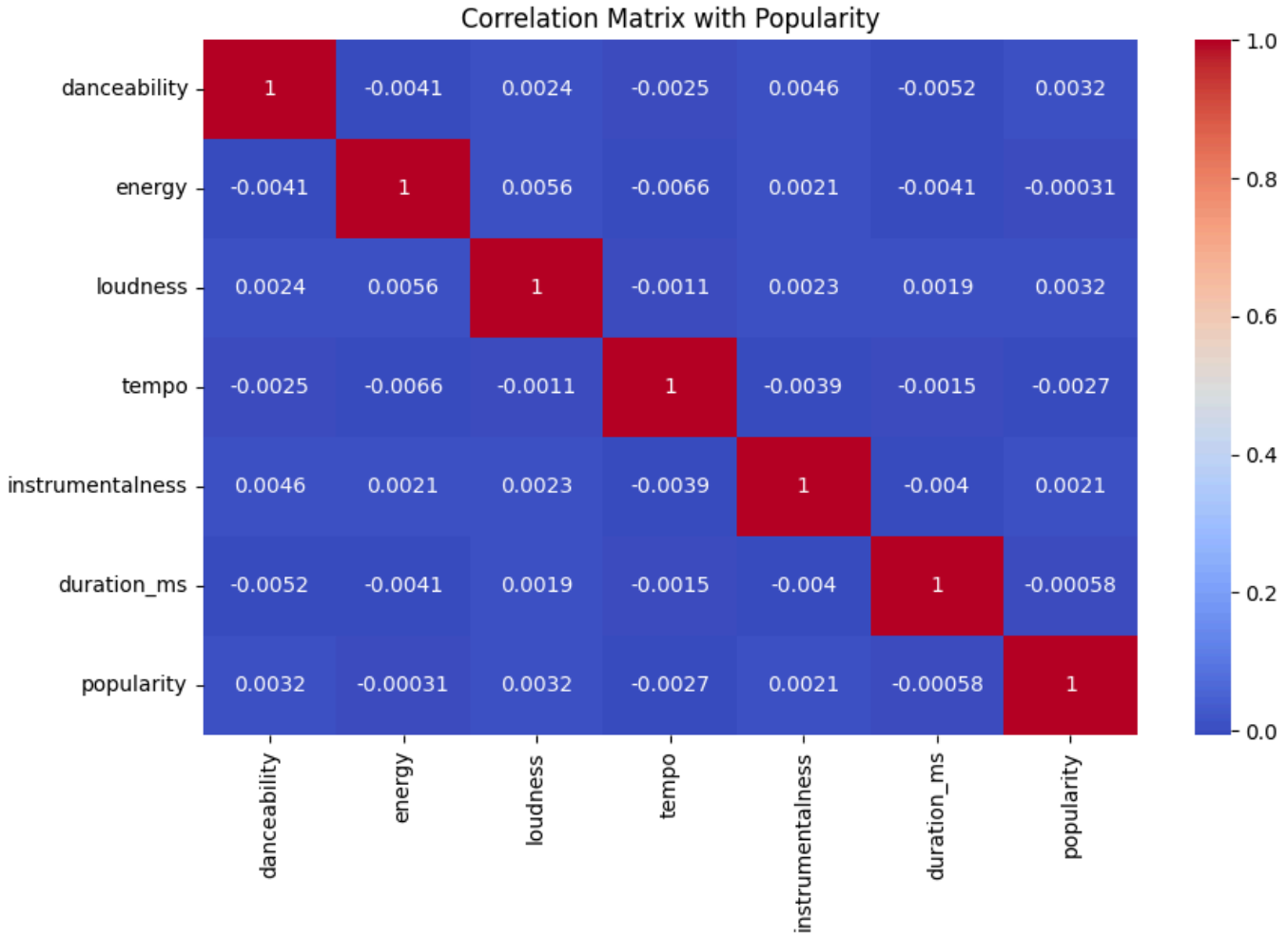
The following chart shows the relative importance of each feature in predicting popularity, as determined by the Random Forest model.





Correlation Matrix

The correlation matrix below illustrates the relationship between various audio features and the target variable (popularity).



3. Insights and Conclusions

Based on the analysis, several key insights can be drawn:

- **Dominance of Stream Count**: The `stream_count` feature is by far the most significant predictor of a track's popularity. This suggests a strong direct relationship between the number of streams a track receives and its overall popularity score.

- **Audio Features Impact**: While features like `duration_ms`, `tempo`, and `loudness` contribute to the model, their individual impact is significantly lower than that of `stream_count`. Core musical attributes such as `danceability` and `energy` showed the least importance in this specific predictive task.
- **Model Selection**: Given that the Linear Regression model performed slightly better and is computationally less expensive than the Random Forest Regressor, it may be the preferred choice for this dataset unless further non-linear relationships are explored.
- **Predictive Limitations**: An R^2 of 0.45 indicates that more than half of the variance in popularity is still unexplained. This suggests that other factors not included in the dataset—such as artist reputation, marketing efforts, social media trends, or genre-specific cultural shifts—likely play a major role in determining a track's success.